

Dragonfishes and relatives

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Stenopterygii
ORDERS	2
FAMILIES	5
SPECIES	About 430

Dragonfishes and their relatives are deep-sea predators with unusual adaptations for catching prey, including large, flexible jaws, sharp teeth, and light-emitting lures. They also have luminous organs on their body known as photophores. This group also includes bristlemouths, viperfishes, slackjaws, and snaggleteooths. Their names may reflect their ferocity, but these are relatively small fishes

whose prey consists of other deep-sea fishes and invertebrates. Some recent analysis suggests that dragonfish may belong to a new superorder group, called Osmeromorpha (see p.501).

Anatomy

Most of the fishes in this group have an elongated body and a relatively large head. The head usually contains modified jaws with a very wide gape. For example, slackjaws have no floor to their mouth. Viperfishes have an extra joint in the head that increases their gape; as the jaw opens, organs such as the heart and gills are pushed backward and downward, out of the way of incoming food. The jaws are usually armed with numerous needlelike teeth that are used to seize prey or to prevent food from escaping. Apart from viperfishes, the fishes in

this group have a generally scaleless body that is colored black or brown to match the darkened waters. However, some species are silvery or even transparent. Many species have light-producing organs called photophores. They share this feature with the lanternfishes (see opposite), although in the dragonfishes the photophores are generally larger and more numerous. They are usually located along the sides and underparts of the body or near the eyes. Each is studlike in shape and linked to

a nerve. Light is reflected through a silvery backing and focused through a lenslike thickening of the overlying scale. They either shine constantly, or flash on and off. Species that live in shallower water—at depths to about 1,000 ft (300 m)—generally have fewer and weaker photophores, and the light output seems to vary according to the amount of light received from above. Thus, photophores may hide the fishes from predators below them by matching the background illumination from above. Almost all species in this group have a light organ at the end of a chin barbel or a specialized fin ray. This is used as a lure to attract prey within striking distance.



FEEDING
Dragonfishes, such as this black dragonfish, are impressive predators. Many species feed near the surface at night and move to deeper water to rest during the day. The body is illuminated by photophores.

Chauliodus sloani

Sloane's viperfish

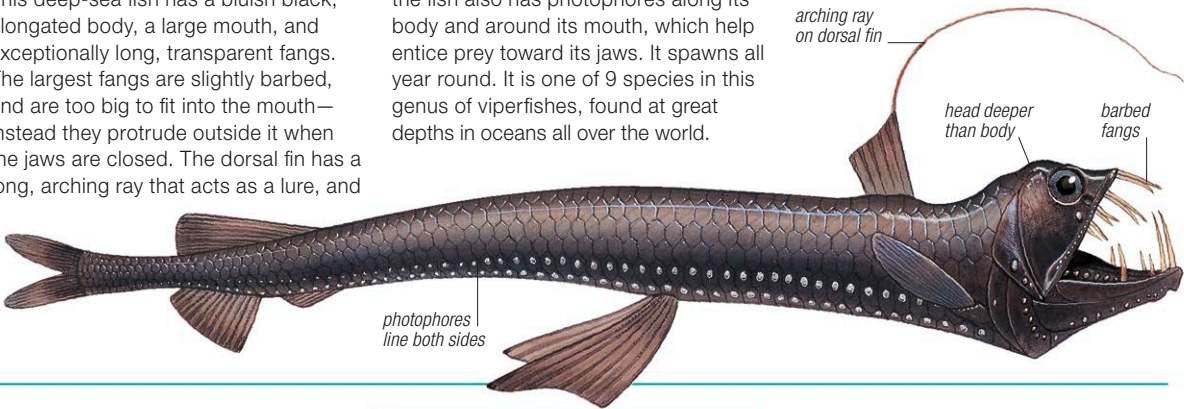


Length Up to 14 in (35 cm)
Weight Up to 1 1/16 oz (30 g)
Sex Male/Female
Status Least concern

Location Tropical, subtropical, and temperate waters worldwide

This deep-sea fish has a bluish black, elongated body, a large mouth, and exceptionally long, transparent fangs. The largest fangs are slightly barbed, and are too big to fit into the mouth—instead they protrude outside it when the jaws are closed. The dorsal fin has a long, arching ray that acts as a lure, and

the fish also has photophores along its body and around its mouth, which help entice prey toward its jaws. It spawns all year round. It is one of 9 species in this genus of viperfishes, found at great depths in oceans all over the world.



Idiacanthus antrostomus

Pacific blackdragon



Length Female up to 15 in (38 cm)
Weight Up to 2 oz (55 g)
Sex Male/Female
Status Not evaluated

Location E. Pacific

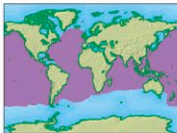
This bizarre, deep-water marine fish has a black, snakelike body and extremely large teeth that must be rotated to allow the mouth to open and close. Females are about 40 times larger than males. To attract prey, the Pacific blackdragon uses a movable light-producing organ located at the end of a barbel that protrudes from the front of the lower jaw. Other photophores run along the belly. Blackdragons feed by night, moving nearer the surface to prey

on small fishes and crustaceans. The larvae are remarkable in having eyes set on long stalks and intestines that extend beyond the tail.



Malacosteus niger

Northern stoplight loosejaw



Length Up to 9 1/2 in (24 cm)
Weight Not recorded
Sex Male/Female
Status Least concern

Location Tropical, subtropical, and temperate waters worldwide

The Northern stoplight loosejaw is unique in having no membranes joining its jaws and tongue—a feature that may help it make a rapid strike at prey. The head can pivot and the jaws are protractile, which allows it to draw in prey—fishes and crustaceans—much larger than itself. While many species in this family have light-producing organs (photophores) arranged in rows along the body, the Northern stoplight loosejaw has them only around its mouth and is the only fish to produce red bioluminescence.

